Biswajyoti Nath

CS Undergrad • Building with AI & Quantum ⚛ | Python • ML • Prompt Engineering | NSP Scholar

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SUMMARY

Computer Science Engineering student specializing in cybersecurity and emerging quantum-safe technologies.
Experience in Python-based ML systems, secure protocol concepts, and applied computer vision.
Published analytical work on randomness in quantum cryptography and building foundations in post-quantum cryptographic systems and distributed security models.

SKILLS

Programming
- Python
- C, C++
- Object-Oriented Programming (OOP)
- Data Structures and Algorithms

Machine Learning
- NumPy
- Scikit-Learn
- Computer Vision (YOLO, OpenCV)
- Optimization Algorithms
- Numerical Optimization
- Statistical Learning

Cybersecurity
- Linux Security
- Network Analysis
- Cryptography Fundamentals
- Ethical Hacking

Tools
- Git & GitHub
- Ubuntu
- VS Code

Conceptual / Technical Skills
- Analytical Skills
- Software Development Best Practices

EDUCATION

Bachelor of Technology - Computer Science & Engineering,
Barak Valley Engineering College (ASTU)
08/2024 – Present | Sribhumi, Assam
- Awarded the Ishan Uday Scholarship (Government of India) for academic merit, providing full financial support during undergraduate studies.

High School Diploma,
Jawahar Navodaya Vidyalaya
09/2022 – 05/2024 | Rangia, Assam

Projects

Automated Cattle Condition Analyzer
08/2025 – 11/2025
Computer vision system built with YOLOv8 and OpenCV to detect cattle, extract geometric body features, and estimate livestock body condition scores through automated image analysis.

Startup LLM Content Pipeline
03/2025
End-to-end AI pipeline that discovers startups, extracts structured insights using large language models, and generates formatted content. Combines automated search, data extraction, and LLM reasoning to transform raw web sources into structured information.

CERTIFICATIONS

Python Essentials 1 - Cisco Networking Academy | Python Institute

IBM Quantum Enigmas (Levels I-II)
Completed advanced quantum logic challenges using circuit abstractions, controlled operations, and measurement-based state elimination.

Python Essentials 2 - Cisco Networking Academy | Python Institute

Machine Learning with Python
Supplementary certification going on alongside independent study of machine learning.

Ethical Hacker - Cisco Networking Academy
Trained in vulnerability assessment and ethical security testing practices.

Research and Publications

- Comparative analysis of pseudorandom vs quantum random number generators in quantum key distribution protocols.
- Implemented reproducible experiments using Qiskit demonstrating QRNG advantages in secure communication.
- Received conceptual clarification from Prof. Gilles Brassard and Prof. Nicolas Gisin regarding protocol assumptions and randomness in quantum cryptography.
- Published with open-access DOI for academic transparency and reproducibility.

INVOLVEMENT

Rotary Club Silchar, Volunteer 07/2025 – 08/2025 | Silchar
- Participated in organizing health campaigns and social welfare events.
- Conducted STEM teaching sessions for students in underserved communities.

Workshop Participant, Encoders Pro Pvt. Ltd. 03/2025 – 03/2025
- Hands-on exposure to core cybersecurity concepts, threat models, and basic security workflows.